

LSEG Quantitative Analytics

Database Schema for Toyo Keizai

Version 1.0.5



LSEG DATA &
ANALYTICS

Contents

1 Introduction.....	3
In This Document.....	3
Intended Readership.....	3
Table Name Conventions	3
Toyo Keizai Version 2 and Legacy Toyo Keizai	3
Pervasive and Oracle	4
Zero Values and Nulls.....	4
Service and Support	4
Feedback	5
Your Personal Information	5
2. TREND³EPOT – Toyo Keizai Version 2.....	6
TK2AdjFactor: Adjustment Factors	7
TK2CumAdjFactor: Cumulative Adjustment Factors.....	7
TK2Catalog: Toyo Keizai 2 Catalogs	7
TK2Code: Toyo Keizai 2 Codes.....	8
TK2Exchange: Listing Exchanges	8
TK2Info: Company Information	8
TK2InfoChg: Company Information Changes	9
TK2Item: Financial Statement Account Items	9
TK2Shares: Historical Shares	10
TK2*Data: Toyo Keizai 2 Data Tables	10
TK2Notes: Estimate Footnotes	12

1 Introduction

In This Document

This document provides schema information about the **Toyo Keizai** data tables in Quantitative Analytics.

NOTE

For information about the other Quantitative Analytics tables, refer to the other Database Schema documents posted on the [Quantitative Analytics](#) product page on [MyAccount](#).

The documentation for each table includes these main elements:

- **Vendor Database or License Name** – This element provides the vendor database or license name. Vendor databases generally correspond to vendor licensing options.
- **Table Name: Table Title** – This element provides the table name and title, followed by a description of the table contents.
- **Update Cycle** – This element indicates the frequency of updates from the data vendor. (For specific information about when this update is posted for your use, refer to your update logs. The update log schema description is available in the Database Schema for Quantitative Analytics Core Tables document.)
- **Adjusted** – This element indicates whether the data is stored adjusted, unadjusted, or if adjustment is not applicable (N/A).
- **Indexes, Index Fields** – This element lists the table's index(es) (for example, pkey_CSGCAD) and the index fields. The primary index contains a "pkey_" or "PK_" prefix. Clustered indexes are noted.
- **Field, Type, Nullable, Definition** – This element describes the table's fields, including field names, data types, and whether each field is nullable. The data types in Snowflake have equivalents available. For example, an *int* in SQL Server is a *number* in Snowflake.

Intended Readership

This document is intended for Quantitative Analytics clients licensed to use **Toyo Keizai** data through LSEG. Note that the tables described in this document may be covered by several licenses; your access to the data in these tables is dependent on the licenses you have. Readers should have basic knowledge of Microsoft SQL Server 2016.

Table Name Conventions

In Quantitative Analytics, database tables are grouped by name. Generally, the first few letters of the table name refer to the data vendor or licensing group. For example, all tables that begin with **TK2*** are **Toyo Keizai Version 2** data tables.

Toyo Keizai Version 2 and Legacy Toyo Keizai

Due to changes in the Toyo Keizai datafeed, Quantitative Analytics release 5.84 contains Toyo Keizai version 2. The Toyo Keizai Version 2 schema contains the TREND3EPOT datafeed and supercedes the Toyo Keizai schema that existed prior to release 5.84. The pre-5.84 schema is now considered "legacy." Tables in the legacy Toyo Keizai schema begin with the letters TK (rather than TK2).

The legacy tables will be discontinued in the future on a date yet to be determined. In the meantime, they remain supported by LSEG Quantitative Analytics. It is recommended that you upgrade to the LSEG Financials schema on LSEG Quantitative Analytics deployed version 5.84 (minimum) to ensure continued support of your data.

Pervasive and Oracle

This document is specific to Microsoft SQL Server 2016. If you are accessing the data directly through the Pervasive database, you may encounter slight differences in index names and data types. Please note that references to Pervasive and Actian are synonymous.

If you are accessing the data through Oracle, note that field names that correspond to Oracle reserved words are appended with an underscore. For example, field names “Number”, “Synonym”, and “UID” become “Number_”, “Synonym_”, and “UID_” in Oracle.

Zero Values and Nulls

Null values are usually recorded as “NULL”. Occasionally, a table may contain a value of “-99999” or “-9999”; in these cases, there is no recorded data for that period. (It is null.) If the table contains a “0” value, the data was reported as having a “0” value.

Service and Support

[MyAccount](#) is the LSEG portal that provides a single access point for timesaving support services, along with billing, user management, and information.

Statement of Service

The [Quantitative Analytics Service Description](#) is available on the [Quantitative Analytics](#) product page at MyAccount.

Subscribe to Support Channels

You are encouraged to [subscribe](#) to the following support channels to keep informed of changes to products and data, and to be notified of any service issues or change. Please see the *Alerts and Notifications* section in the *Quantitative Analytics Service Description* available on the [Quantitative Analytics](#) product page at MyAccount for detailed instructions.

- **Change Notifications**
 - **Product Change Notifications (PCNs)** detail new, enhanced, or changed functionality, which may require your action, in products that you use.
 - **Data Notifications (DNs)** alert you to upcoming changes to real-time and historical data across all asset classes that are relevant to you.
 - **RIC Change Events** inform you of planned changes to RICs. RIC is the market level identifier key unique to LSEG used to identify securities.
- **Service Alerts**

Alert you about planned maintenance and unplanned service issues affecting your products and services, and be notified via SMS or email.

Documentation

You can access product documentation on the [Quantitative Analytics](#) product page at MyAccount.

Get Support

For support using Quantitative Analytics, please use one of the following methods:

Online

Access [Get Support](#) either via the blue [Get Support](#) button at the top right of each page or by selecting *Get Support* located under *Help and Support* on the left navigation pane at MyAccount.

From the *Get Support* page, choose *Get Product and Content Support* and then select the appropriate category from the next page. Complete the request making sure to include the following:

- Under Product, select Quantitative Analytics from the dropdown list.
- In the **How can we help you?** text box, include the following information:

- Content set name (i.e. Worldscope)
- Date applicable to a content inquiry
- Table names and relevant fields
- Security code (i.e. QA-ID, PermID, Sedol, InfoCode, etc.)
- Script used to generate the output
- Any other supporting detail

Telephone

Region	Telephone	Options
AMERS	+01 866 427 8390	Option 1 – Product Support Option 2 – Technical Support
EMEA	+44 203 341 1036	
APAC	+65 6403 7436	

Feedback

We invite your comments, corrections, and suggestions about this document: access the [Feedback](#) option under [Get Support](#) at MyAccount. Your feedback helps us continue to improve our user assistance.

Your Personal Information

LSEG is committed to the responsible handling and protection of personal information. We invite you to review our [Privacy Statement](#), which describes how we collect, use, disclose, transfer, and store personal information when needed to provide our services and for our operational and business purposes.

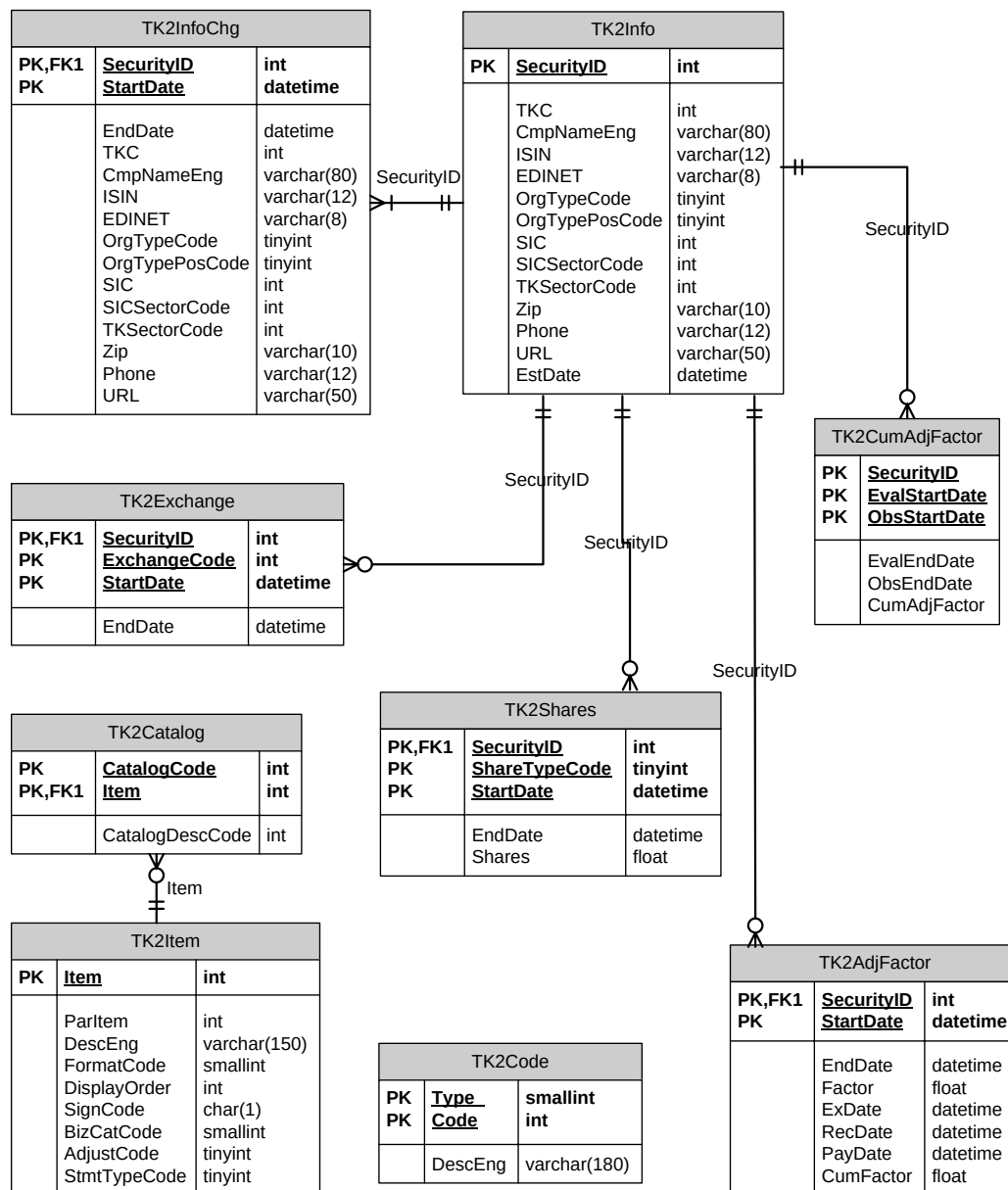
2. TREND³EPOT – Toyo Keizai Version 2

The LSEG Quantitative Analytics deployed database schema for Toyo Keizai Version 2 contains the Toyo Keizai TREND³EPOT data feed for fundamentals and earning estimates data.

The tables described in this section are available in both LSEG Quantitative Analytics deployed and in the cloud.

Update Cycle: Daily

Adjusted: Yes, where applicable



TK2AdjFactor: Adjustment Factors

Use this table to retrieve historical adjustment factors for per share data items.

Indexes		Index Fields	
pkey_TK2AdjFactor (clustered)		SecurityID, StartDate	

Field	Type	Nullable	Description
SecurityID	int	N	SecurityID is a unique ID number assigned by Toyo Keizai to each company covered. If a company has more than one listing of common stock, each has a unique ID.
StartDate	datetime	N	StartDate is the effective start date for the adjustment factor.
EndDate	datetime	Y	EndDate is the effective end date for the adjustment factor.
Factor	float	Y	Factor is the adjustment factor used for stock prices at the event. Events may be stock splits, paid share allotments, capital decreases, or reverse stock splits.
ExDate	datetime	Y	ExDate is the ex-rights date of the capital action.
RecDate	datetime	Y	RecDate is the record date of the capital action.
PayDate	datetime	Y	PayDate is the payment date of the capital action.
CumFactor	float	Y	CumFactor is the LSEG-calculated cumulative adjustment factor for historical per share data. CumFactor is derived by LSEG using the following formula: IF current record (that is, if EndDate is NULL), CumFactor = 1 ELSE, CumFactor = (Prev CumFactor / Prev Factor)

TK2CumAdjFactor: Cumulative Adjustment Factors

Use this table to retrieve historical, cumulative adjustment factors for per share data items.

Indexes		Index Fields	
pkey_TK2CumAdjFactor (clustered)		SecurityID, EvalStartDate, ObsStartDate	

Field	Type	Nullable	Description
SecurityID	int	N	SecurityID is a unique ID number assigned by Toyo Keizai to each company covered. If a company has more than one listing of common stock, each has a unique ID.
EvalStartDate	datetime	N	The evaluation start date; it is a point-in-time representation of the adjustment factor. A single evaluation point can have multiple observation points.
ObsStartDate	datetime	N	The observation start date; the observation point falls between the observation start and end date. This represents the point in time in which the security is being observed; when backtesting historical observation points should be used.
EvalEndDate	datetime	Y	The evaluation end date; it is a point-in-time representation of the adjustment factor. A single evaluation point can have multiple observation points.
ObsEndDate	datetime	Y	The observation end date; the current observation period is represented by ObsEndDate = 1999-12-31.
CumAdjFactor	float	Y	The cumulative adjustment factor.

TK2Catalog: Toyo Keizai 2 Catalogs

This table contains the names of the data package services provided by Toyo Keizai and the items that belong to each catalog.

Indexes		Index Fields	
pkey_TK2Catalog (clustered)		CatalogCode, Item	

Field	Type	Nullable	Description
CatalogCode	int	N	CatalogCode is the unique identifier for each catalog. TK2Catalog.CatalogCode cross-references with TK2Code.Code where Type_ = 12.
Item	int	N	Item is the unique item ID. Item cross-references TK2Item .
CatalogDescCode	int	Y	CatalogDescCode is the catalog description code. TK2Catalog.CatalogDescCode cross-references with TK2Code.Code where Type_ = 13.

TK2Code: Toyo Keizai 2 Codes

This table contains descriptions for Toyo Keizai coded items.

Indexes		Index Fields	
pkey_TK2Code (clustered)		Type_, Code	
Field	Type	Nullable	Description
Type_	smallint	N	Type_ identifies the code category. Types are defined where Type_ = 0 and Code identifies Type_.
Code	int	N	Code is described by DescEng. Code is cross-referenced from other tables.
DescEng	varchar(180)	Y	DescEng is the full name or description of Code in English.

TK2Exchange: Listing Exchanges

Use this table to retrieve the exchange on which each company is listed.

Indexes		Index Fields	
pkey_TK2Exchange (clustered)		SecurityID, ExchangeCode, StartDate	
Field	Type	Nullable	Description
SecurityID	int	N	SecurityID is a unique ID number assigned by Toyo Keizai to each company covered. If a company has more than one listing of common stock, each has a unique ID.
ExchangeCode	int	N	ExchangeCode indicates the exchange on which the company trades. TK2Exchange.ExchangeCode cross-references with TK2Code .Code where Type_ = 16.
StartDate	datetime	N	StartDate is the first listing date.
EndDate	datetime	Y	EndDate is the last listing date. If the item is active, EndDate is 9999-12-31

TK2Info: Company Information

This table lists the companies covered by this implementation of the Toyo Keizai TREND³EPOT datafeed.

Indexes		Index Fields	
pkey_TK2Info (clustered)		SecurityID	
Field	Type	Nullable	Description
SecurityID	Int	N	SecurityID is a unique ID number assigned by Toyo Keizai to each company covered. If a company has more than one listing of common stock, each has a unique ID.
TKC	Int	Y	TKC indicates the original Toyo Keizai company code.
CmpNameEng	varchar(80)	Y	CmpNameEng is the company name in English.
ISIN	varchar(12)	Y	ISIN is the company International Securities Identification Number (ISIN).
EDINET	varchar(8)	Y	EDINET is the company identifier assigned by the Electronic Disclosure for Investors NETwork (EDINET).
OrgTypeCode	tinyint	Y	OrgTypeCode identifies the type of organization. TK2Info.OrgTypeCode cross-references with TK2Code .Code where Type_ = 1.
OrgTypePosCode	tinyint	Y	OrgTypePosCode identifies the position in the company name of the organization type. For example, if the company name is Sanyo Corporation, then this code indicates that Corporation comes after the company name, Sanyo. TK2Info.OrgTypePosCode cross-references with TK2Code .Code where Type_ = 2.
SIC	int	Y	SIC stands for Securities Identification Code.
SICSectorCode	int	Y	SICSectorCode is a three-digit code that indicates the company sector. TK2Info.SICSectorCode cross-references with TK2Code .Code where Type_ = 17.
TKSectorCode	int	Y	TKSectorCode indicates the company's fourth-level sector classification based on Toyo Keizai's original sector hierarchy.

Field	Type	Nullable	Description
			TK2Info.TKSectorCode cross-references with TK2Code .Code where Type_ = 18.
Zip	varchar(10)	Y	Zip is the company's zip code.
Phone	varchar(12)	Y	Phone is the company's phone number.
URL	varchar(50)	Y	URL is the company's Web site address.
EstDate	datetime	Y	EstDate is the date of establishment of the company.

TK2InfoChg: Company Information Changes

This table contains compiled historical changes of any of several identifiers in the TREND³EPOT datafeed.

Indexes	Index Fields
pkey_TK2InfoChg (clustered)	SecurityID, StartDate

Field	Type	Nullable	Description
SecurityID	int	N	SecurityID is a unique ID number assigned by Toyo Keizai to each company covered. If a company has more than one listing of common stock, each has a unique ID.
StartDate	datetime	N	StartDate is the valid start date for the element, such as company name or ISIN.
EndDate	datetime	Y	End Date is the valid end date for the element, such as company name or ISIN.
TKC	int	Y	TKC indicates the original Toyo Keizai company code.
CmpNameEng	varchar(80)	Y	CmpNameEng is the company name in English.
ISIN	varchar(12)	Y	ISIN is the company International Securities Identification Number (ISIN).
EDINET	varchar(8)	Y	EDINET is the company identifier assigned by the Electronic Disclosure for Investors NETwork (EDINET).
OrgTypeCode	tinyint	Y	OrgTypeCode identifies the type of organization. TK2InfoChg.OrgTypeCode cross-references with TK2Code .Code where Type_ = 1.
OrgTypePosCode	tinyint	Y	OrgTypePosCode identifies the position in the company name of the organization type. For example, if the company name is Sanyo Corporation, then this code indicates that Corporation comes after the company name, Sanyo. TK2InfoChg.OrgTypePosCode cross-references with TK2Code .Code where Type_ = 2.
SIC	int	Y	SIC stands for Securities Identification Code.
SICSectorCode	int	Y	SICSectorCode is a three-digit code that indicates the company sector. TK2InfoChg.SICSectorCode cross-references with TK2Code .Code where Type_ = 17.
TKSectorCode	int	Y	TKSectorCode indicates the company's fourth-level sector classification based on Toyo Keizai's original sector hierarchy. TK2InfoChg.TKSectorCode cross-references with TK2Code .Code where Type_ = 18.
Zip	varchar(10)	Y	Zip is the company's zip code.
Phone	varchar(12)	Y	Phone is the company's phone number.
URL	varchar(50)	Y	URL is the company's Web site address.

TK2Item: Financial Statement Account Items

The data in this table identifies all the account items of financial statements referenced in the TREND³EPOT datafeed.

Indexes	Index Fields
pkey_TK2Item (clustered)	Item

Field	Type	Nullable	Description
Item	int	N	Item is the unique item ID. This ID is cross-referenced from other tables.

Field	Type	Nullable	Description
ParItem	int	Y	ParItem identifies the parent account item, if applicable.
DescEng	varchar(150)	Y	DescEng is the full name or description of Item in English.
FormatCode	smallint	Y	FormatCode indicates the datatype of the item. TK2Item.FormatCode cross-references with TK2Code .Code where Type_ = 10.
DisplayOrder	int	Y	DisplayOrder indicates the presentation order of each item.
SignCode	char(1)	Y	SignCode indicates how each account item is calculated in relation to its parent item. TK2Item.SignCode cross-references with TK2Code .Code where Type_ = 11.
BizCatCode	smallint	Y	BizCatCode indicates the business category for each item. TK2Item.BizCatCode cross-references with TK2Code .Code where Type_ = 9.
AdjustCode	tinyint	Y	(Reserved for future development.)
StmtTypeCode	tinyint	Y	StmtTypeCode identifies the source statement of the item. TK2Item.StmtTypeCode cross-references with TK2Code .Code where Type_ = 20.

TK2Shares: Historical Shares

This table contains historical shares data.

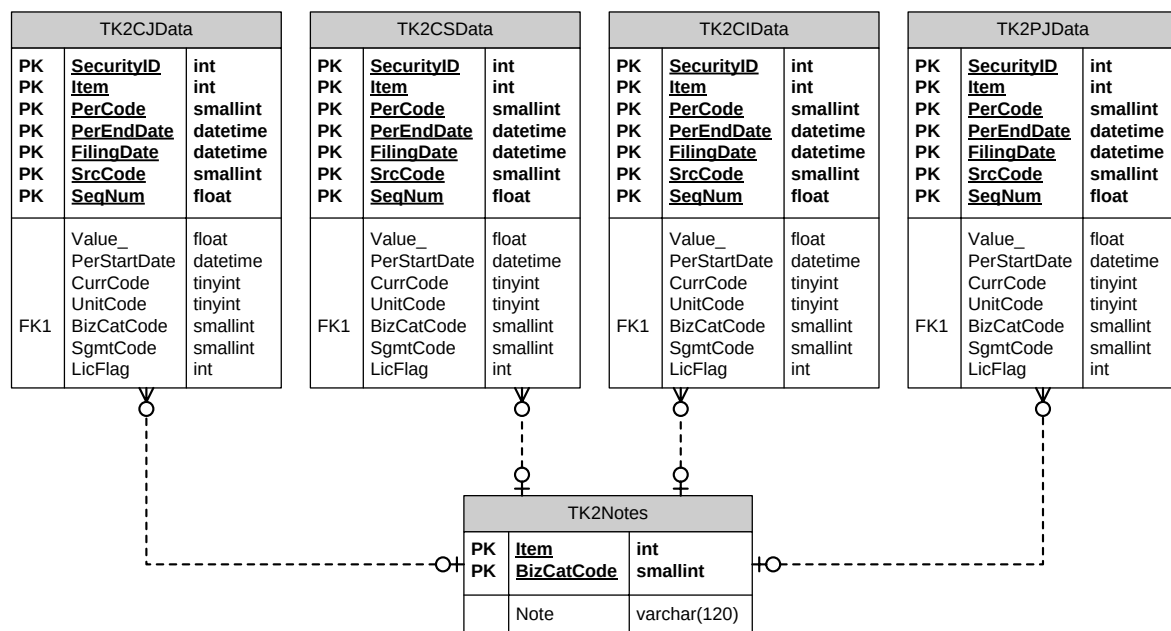
Indexes		Index Fields	
pkey_TK2Shares (clustered)		SecurityID, ShareTypeCode, StartDate	

Field	Type	Nullable	Description
SecurityID	int	N	SecurityID is a unique ID number assigned by Toyo Keizai to each company covered. If a company has more than one listing of common stock, each has a unique ID.
ShareTypeCode	tinyint	N	ShareTypeCode indicates the type base for the share. TK2Shares.ShareTypeCode cross-references with TK2Code .Code where Type_ = 21.
StartDate	datetime	N	StartDate is the effective start date for the shares value.
EndDate	datetime	Y	EndDate is the effective end date for the shares value.
Shares	float	Y	Shares indicates the number of shares outstanding.

TK2*Data: Toyo Keizai 2 Data Tables

This section describes the Toyo Keizai data tables. For these tables, the schema is identical, only the content is different.

- **TK2CJData:** Consolidated Data – Japanese Standards
This table contains data item values and data source information for consolidated data reported under Japanese standards.
- **TK2CSDData:** Consolidated Data – SEC Standards
This table contains data item values and data source information for consolidated data reported under SEC standards.
- **TK2CIDData:** Consolidated Data – IFRS Standards
This table contains data item values and data source information for consolidated data reported under IFRS standards.
- **TK2PJData:** Parent Data – Japanese Standards
This table contains all data item values and source information for parent data reported under Japanese standards.



The TK2*Data tables also link to:

- [TK2Info](#) on SecurityID
- [TK2Item](#) on Item
- [TK2AdjFactor](#) on SecurityID
- [TK2Shares](#) on SecurityID

Indexes	Index Fields
pkey_TK*Data (clustered)	SecurityID, Item, PerCode, PerEndDate, FilingDate, SrcCode, SeqNum

Field	Type	Nullable	Description
SecurityID	int	N	SecurityID is a unique ID number assigned by Toyo Keizai to each company covered. If a company has more than one listing of common stock, each has a unique ID.
Item	int	N	Item is the unique item ID. Item cross-references TK2Item .
PerCode	smallint	N	PerCode indicates the type of period. TK2*Data.PerCode cross-references with TK2Code .Code where Type_ = 3.
PerEndDate	datetime	N	PerEndDate is the period end date.
FilingDate	datetime	N	FilingDate is the date the source documents were filed. Note that for Yuho reports published before March 1992, the filing date may not be available. In this case, the date and time provided is: 1900-01-01 00:00:00.
SrcCode	smallint	N	SrcCode indicates the source document type. TK2*Data.SrcCode cross-references with TK2Code .Code where Type_ = 6.
SeqNum	float	N	SeqNum indicates the order of the company segment.
Value_	float	Y	Value_ is the value specified for Item by the SrcCode document.
PerStartDate	datetime	Y	PerStartDate is the period start date.
CurrCode	tinyint	Y	CurrCode indicates the reporting currency. TK2*Data.CurrCode cross-references with TK2Code .Code where Type_ = 7.
UnitCode	tinyint	Y	UnitCode indicates the unit of measure for Value_. TK2*Data.UnitCode cross-references with TK2Code .Code where Type_ = 8.
BizCatCode	smallint	Y	BizCatCode indicates the business category of the company. TK2*Data.BizCatCode cross-references with TK2Code .Code where Type_ = 9.
SgmtCode	smallint	Y	SgmtCode is the segment code. It indicates the reporting company segment.

Field	Type	Nullable	Description
LicFlag	int	Y	LicFlag indicates the licensing for each data point. TK2*Data.LicFlag cross-references with TK2Code .Code where Type_ = 19.

TK2Notes: Estimate Footnotes

Use this table to retrieve footnotes for the estimate items in the TREND³EPOT datafeed.

Indexes	Index Fields
pkey_TK2Notes (clustered)	Item, BizCatCode

Field	Type	Nullable	Description
Item	int	N	Item is the unique item ID. Item cross-references TK2Item .
BizCatCode	smallint	N	BizCatCode indicates the business category for each item. Cross-reference BizCatCode with TK2Code where BizCatCode = TK2Code.Code and TK2Code.Type_ = 9.
Note	varchar(120)	Y	Note is the text of the footnote.

Copyright © 2024 London Stock Exchange Group plc and its group of companies (LSEG) and/or its affiliates. All rights reserved. The LSEG content received through this service is the intellectual property of LSEG or its third party suppliers. Republication or redistribution of content provided by LSEG is expressly prohibited without the prior written consent of LSEG, except where permitted by the terms of the relevant LSEG service agreement. Neither LSEG nor its third party suppliers shall be liable for any errors, omissions or delays in content, or for any actions taken in reliance thereon. LSEG and its logo are trademarks of LSEG.

For more information visit us at www.lseg.com



LSEG DATA &
ANALYTICS